

1. Outcomes, events and random variables

The **sample space** Ω contains the possible outcomes of an experiment. An **event** $A \subseteq \Omega$ is a set of outcomes. A **random variable** assigns a number to each outcome:

$$X : \Omega \longrightarrow \mathbb{R}.$$

Use X for the random variable and x for a realised value. A random vector $\mathbf{X} = (X_1, \dots, X_d)^\top$ records several quantities.

Example: two coin flips. For $\Omega = \{HH, HT, TH, TT\}$, the number of heads satisfies $X(HH) = 2$, $X(HT) = X(TH) = 1$, $X(TT) = 0$. Photon counts are discrete; a temperature is modelled as continuous.

Kolmogorov axioms. For any event A , and any countable collection of pairwise disjoint events A_i ,

$$\mathbb{P}(A) \geq 0, \quad \mathbb{P}(\Omega) = 1, \quad \mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

Consequently $0 \leq \mathbb{P}(A) \leq 1$ and $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Interpretation. Frequentist probability is linked to relative frequencies in repeated experiments under the same conditions. Finite sequences can give different frequencies: the observed fraction is an estimate, not an exact probability. The axioms supply the probability calculus shared by frequentist and Bayesian inference.

2. Mass functions and density functions

Discrete: probability mass function (PMF). For a finite or countable support $\mathcal{S}$,

$$p_X(x) = \mathbb{P}(X = x), \quad p_X(x) \geq 0, \\ \sum_{x \in \mathcal{S}} p_X(x) = 1, \quad \mathbb{P}(X \in A) = \sum_{x \in A \cap \mathcal{S}} p_X(x).$$

For two independent fair coin flips, the head-count PMF is $p_X(0) = 1/4$, $p_X(1) = 1/2$, $p_X(2) = 1/4$.

Continuous: probability density function (PDF). For a variable with density p_X ,

$$p_X(x) \geq 0, \quad \int_{-\infty}^{\infty} p_X(x) dx = 1, \\ \mathbb{P}(a \leq X \leq b) = \int_a^b p_X(x) dx.$$

Here $\mathbb{P}(X = x) = 0$ at every individual point. For a small interval where the density varies little,

$$\mathbb{P}(x \leq X \leq x + \Delta x) \approx p_X(x) \Delta x.$$

Density is not probability: a PDF can exceed one. Its units are inverse units of x ; probability is the dimensionless area under the curve. A PMF gives probability directly at each support value.

3. Cumulative distribution functions

The **CDF**, $F_X(x) = \mathbb{P}(X \leq x)$, applies to both discrete and continuous variables. It is non-decreasing and right-continuous, with limits 0 and 1 at $-\infty$ and $+\infty$. For any $a < b$,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

With a density: integration gives the CDF; differentiation recovers the density almost everywhere:

$$F_X(x) = \int_{-\infty}^x p_X(t) dt, \quad p_X(x) = \frac{dF_X(x)}{dx}.$$

Discrete: the CDF accumulates masses, and each jump is a mass:

$$F_X(x) = \sum_{u \leq x} p_X(u), \quad p_X(x) = F_X(x) - F_X(x^-).$$

Here $F_X(x^-)$ is the left limit. Probability is a *vertical difference* on a CDF plot, not the area beneath it.

4. Joint, marginal and conditional distributions

A **joint density** $p_{X,Y}(x,y)$ describes two variables together. It is nonnegative and integrates to one over both variables. **Marginalise** to ignore an unobserved variable: integrate over all its possible values,

$$p_X(x) = \int p_{X,Y}(x,y) dy.$$

Condition to describe X at a specified $Y = y$. For $p_Y(y) > 0$,

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}, \quad \int p_{X|Y}(x|y) dx = 1.$$

Thus the joint distribution factors in either order:

$$p_{X,Y}(x,y) = p_{X|Y}(x|y)p_Y(y) \\ = p_{Y|X}(y|x)p_X(x).$$

For discrete variables, replace integrals by sums. Marginalisation averages over the other variable; conditioning selects and normalises a slice of the joint density.

5. Change of variables

Probability is conserved when coordinates change. If $Y = g(X)$ is a differentiable, invertible transformation with nonzero derivative,

$$p_Y(y) = p_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|.$$

For a smooth invertible map $\mathbf{y} = g(\mathbf{x})$ in d dimensions with nonsingular Jacobian,

$$p_{\mathbf{Y}}(\mathbf{y}) = p_{\mathbf{X}}(\mathbf{x}) \left| \det \frac{\partial \mathbf{x}}{\partial \mathbf{y}} \right|, \quad \mathbf{x} = g^{-1}(\mathbf{y}).$$

The absolute Jacobian determinant accounts for volume changes. Such density transformations underlie normalising flows.

Lecture 1: Probability foundations

Bayesian & Frequentist Probability

ODSL / LMU

Moments, repeated sampling and Monte Carlo

Summary sheet 2 of 2

6. Expectation, variance and moments

The **expectation** is a probability-weighted average. For any function h whose expectation exists,

$$\mathbb{E}[h(X)] = \begin{cases} \sum h(x)p_X(x), & \text{discrete,} \\ \int h(x)p_X(x) dx, & \text{continuous.} \end{cases}$$

The **mean** and **variance** describe location and squared spread:

$$\mu = \mathbb{E}[X], \\ \sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2.$$

The standard deviation σ has the same units as X . Useful rules, when the moments exist, are

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c, \\ \text{Var}(aX + b) = a^2 \text{Var}(X).$$

Linearity of expectation does not require independence.

The k th **central moment** is $m_k = \mathbb{E}[(X - \mu)^k]$. Here $m_1 = 0$ and $m_2 = \sigma^2$; higher moments describe features such as asymmetry and tail behaviour. Different distributions can have the same mean and variance. For a Gaussian distribution, the mean and covariance determine the entire distribution.

7. Covariance, correlation and independence

Covariance describes how two variables fluctuate together:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

For a vector with mean $\mu = \mathbb{E}[\mathbf{X}]$, its **covariance matrix** is

$$\Sigma = \mathbb{E}[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^T], \\ \Sigma_{ij} = \text{Cov}(X_i, X_j), \quad \Sigma_{ii} = \text{Var}(X_i).$$

It is symmetric and positive semidefinite. In particular,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

Correlation is a dimensionless measure of linear association, defined for finite, nonzero variances:

$$\rho_{XY} = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

Independence requires factorisation of the full joint distribution:

$$p_{X,Y}(x, y) = p_X(x)p_Y(y).$$

Then $p_{X|Y}(x | y) = p_X(x)$ wherever the conditional density is defined. Independence implies zero covariance when the variances are finite; **zero correlation does not imply independence**.

Example. If $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$, then $\text{Cov}(X, Y) = \mathbb{E}[X^3] = 0$, although Y is determined by X . For *jointly Gaussian* variables, zero covariance does imply independence.

8. Sample means and the law of large numbers

Let $X_1, \dots, X_n$ be **independent and identically distributed** (i.i.d.), with mean μ and finite variance σ^2 . Then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

The **sampling distribution** of $\bar{X}_n$ is the distribution of means from repeated datasets, each of size n .

The **weak law of large numbers (LLN)** states that

$$\bar{X}_n \xrightarrow{p} \mu, \quad \text{i.e.} \quad \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \rightarrow 0$$

for every $\varepsilon > 0$. The sample mean concentrates around the population mean. This does not require monotonic improvement along one sequence of observations.

9. Central limit theorem

Under the same i.i.d. assumptions, with $0 < \sigma^2 < \infty$,

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

Thus for sufficiently large n , the sampling distribution is approximately

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \quad \text{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

The approximation is exact for every n if the observations are Gaussian. In general, its accuracy depends on the original distribution.

Distinguish the limits: the LLN describes concentration of $\bar{X}_n$; the CLT describes the shape of its *rescaled fluctuations*. The observations themselves need not become Gaussian.

10. Monte Carlo estimation

Turn an integral into an expectation, draw i.i.d. $X_i \sim p_X$, and average:

$$I = \int f(x)p_X(x) dx = \mathbb{E}[f(X)], \\ \hat{I}_n = \frac{1}{n} \sum_{i=1}^n f(X_i), \quad \text{SE}(\hat{I}_n) = \sqrt{\frac{\text{Var}(f(X))}{n}}.$$

For $\mathbb{E}|f(X)| < \infty$, the LLN gives convergence. Finite variance gives the CLT error scale $n^{-1/2}$: halving it requires four times as many independent draws.

Example: estimating π . Draw independent points (U_i, V_i) uniformly in $[0, 1]^2$. The unit quarter-circle has area $\pi/4$. With $A_i = \mathbf{1}\{U_i^2 + V_i^2 \leq 1\}$,

$$\mathbb{E}[A_i] = \frac{\pi}{4}, \quad \hat{\pi}_n = \frac{4}{n} \sum_{i=1}^n A_i, \quad \text{Var}(\hat{\pi}_n) = \frac{\pi(4 - \pi)}{n}.$$

A running estimate shows one experiment as it grows. Repeating the whole experiment at a fixed n reveals the estimator's sampling distribution.